

Ashok Kolipaka

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Data Science and AI professional with 10+ years of experience in machine learning, NLP and AI-driven solutions, specializing in predictive analytics, automation and LLM-powered applications.

Summary

- Proficient in collaborating with stakeholders to translate business problems into actionable data science and AI solutions, leveraging machine learning, and Generative AI techniques, including agentic workflows.
- Skilled in conducting comprehensive exploratory data analysis (EDA) to understand data and identify patterns.
- Experienced in designing, developing, and deploying end-to-end ML and AI solutions, including LLMs, Generative AI models and ML algorithms, ensuring scalability and real-world applicability.
- Adept at building and integrating cutting-edge LLM-powered applications using Retrieval-Augmented Generation (RAG), prompt engineering, fine-tuning, and vector databases to enhance model accuracy and minimize hallucinations.
- Strong background in MLOps, including developing and automating CI/CD pipelines for ML models, ensuring seamless integration, deployment, and monitoring in production environments.
- Capable of creating insightful data visualizations and dashboards to effectively communicate findings to clients.
- Proven ability to implement AI-driven automation solutions, including AI agents, chatbots, and recommendation engines, to enhance operational efficiency and user experience.
- Skilled in creating end to end ML solutions using cloud-based ML platforms GCP and AWS.
- Extensive experience in leading and working within agile development environments.
- Proven ability to deliver AI and ML solutions across sales, finance, retail, manufacturing, and marketing domains.

Technical Skills

- **Machine Learning:** Classification & Regression (Linear Regression, Logistic Regression, Naïve Bayes, Random Forest), Clustering (K-means, DBSCAN, hierarchical clustering), Deep Learning (ANN, RNN, CNN), Dimensionality Reduction (PCA), Natural Language Processing (BOW, Word2Vec, Doc2Vec and Pretrained SBERT), Recommendation Engine, Large Language Models (LLMs), Chatbot, Generative AI, AI Agents, Conversational AI
- **Languages:** SQL, Python (Django, Flask & ML libraries like scikit-learn, pandas, NumPy, PyTorch, Keras, TensorFlow)
- **Frameworks & Tools:** LangChain (Prompt Engineering, Chains, Document Loaders), LangGraph, Agentic Workflows, RAG, Kubernetes, Docker, Jenkins, GitHub, Bitbucket, KubeFlow, JIRA, Airflow, Pyspark, Databricks
- **Cloud & Big Data:** Google Cloud Platform (BigQuery, Dialogflow CX, Document AI, Vertex AI, Cloud Functions, Pub/Sub, Vertex AI Agent Builder), Amazon Web Services (SageMaker, S3, CloudWatch, Redshift, EC2, Lambda, bedrock), Hive, SQL Server, Teradata, Oracle, PostgreSQL, Pinecone, Cassandra
- **Data Visualization:** Tableau, Power BI, QlikView, Looker Studio

Work Experience

JPMorgan Chase & Co

Senior ML Engineer

Dec 2022 – Present

- **VSI Analytics:** Developed an AI-driven web application for error detection and order ETA prediction using clustering and regression ML techniques. Integrated an LLM-powered chatbot with AI Agents for assisting the SREs.
- **Capacity Planning Analytics:** Built an ML solution using forecasting models to predict high-volume transaction days for digital applications, enabling proactive resource allocation.

Responsibilities:

- Design and implement agentic workflows using LangChain and RAG framework.
- Develop pipelines for data transformation and training ML models & deploy to Kubernetes platform.
- Lead a five-member team, overseeing ML architecture and MLOps strategies for model lifecycle management
- Develop Django REST APIs to serve ML models and expose results via web applications.
- Build and optimize CI/CD pipelines for seamless model deployment and monitoring.

- Create Tableau dashboards to present key business insights and model results.

Technologies: Python, NLP, LLMs, Oracle, GCP Vertex AI Agent Builder, Prophet, Docker, SageMaker, S3 bucket, SQL Server

Quantiphi Analytics Limited

Senior ML Engineer

March 2022 – Dec 2022

Client: Logitech

- **Logo Detection and Text extraction:** Developed an AI-powered solution for identifying logos and extracting associated text from design documents for a manufacturing client

Responsibilities:

- Train and fine-tune ML models for computer vision and text extraction using cloud services.
- Develop automated pipelines for efficient model deployment and monitoring.
- Manage client relations, gather requirements, and document project specifications.

Technologies: Python, GCP (AutoML Vision, Entity Extraction, Doc AI, Cloud Storage, Cloud Functions, Pub/Sub, Cloud Run)

Enquero Global Private Limited

Lead Data Scientist

Nov 2018 – March 2022

Client: Adidas, Deutsche Bank, Atlassian

- **Next Best Engagement:** Developed a recommendation engine to personalize user engagement for an e-commerce client, increasing customer retention and revenue.
- **Next Gen Support Assistance:** Created an AI chatbot for users and a web application for IT support teams, the implemented solutions were predicted to reduce the human toil by 30-40% and MTTR by 20-40%.
- **StatusPage Pricing Analytics:** Built a classification model to create the cohorts/classes based on the product subscription description using ML techniques for better pricing plans.

Responsibilities:

- Train and fine-tune NLP models using Python for improved predictive accuracy.
- Experiment with multiple ML models to determine the best fit for business use cases.
- Lead a team of data scientists to develop and deploy AI-driven solutions.
- Develop Flask-based REST APIs to serve ML models in production environments.
- Build interactive Tableau dashboards for data-driven decision-making.

Technologies: Python, MAB, GCP (Big Query, Dialogflow CX, Looker Studio, Cloud Run), Tableau, SageMaker, S3 bucket.

CA Technologies

Data Scientist

May 2015 – Nov 2018

- **Propensity Analytics:** Conceptualized and implemented one stop solution to find high propensity customers, improving sales and marketing targeting strategies
- **Sales 360 Analytics:** Developed a guided analytics system that provided sales insights, featured in Forbes 2016 for its innovation.

Responsibilities:

- Develop predictive analytics models to enhance sales and marketing insights
- Build automated dashboards to track key performance indicators (KPIs).

Technologies: Python, QlikView, Tableau, Power BI, Teradata, Jupyter Notebooks

Education

B.V. Raju Institute of Technology

Aug 2011 - Apr 2015

Bachelor's Technology in Computer Science and Engineering

Certifications

- Cambridge Business English Certified **ID number:** LSO476213 **Ref No:** 173IN1540041
- Fundamentals of Deep Learning from Analytics Vidhya **Credential ID:** b2h8prsucm
- Python Programming from Analytics Vidhya **Credential ID:** vb5cutgk0z
- Certified Python Programmer from HackerRank **Credential ID:** a11b9362ddde